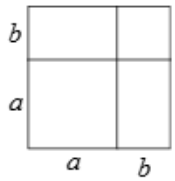
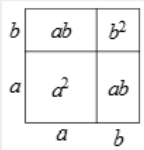
1. Use the figure below to prove that $(a + b)^2 = a^2 + 2ab + b^2$, without expanding algebraically.



Solution: The whole figure is a square with side length $a + b$, so its area is $(a + b)^2$. We can also find the area of the square by adding up the four pieces that together make up the square. The bottom left piece is a square with side length a, so its area is a^2. The upper left and bottom right pieces are each a by b rectangles, so each has area ab. The upper right piece is a square with side length b so its area is b^2. The area of each piece is shown in the diagram at right. The sum of the areas of the pieces is



$$a^2 + ab + ab + b^2 = a^2 + 2ab + b^2.$$

Since these pieces together make up the entire big square, and the big square has area $(a + b)^2$, we have $(a + b)^2 = a^2 + 2ab + b^2$.

2. Compute 31^2 and 299^2 in your head.

Solution: As described in the text, we use the expansion $(a + b)^2 = a^2 + 2ab + b^2$ to perform both computations. First, we let $a = 30$ and $b = 1$ to compute 31^2:

$$31^2 = (30 + 1)^2 = 30^2 + 2(30)(1) + 1^2 = 900 + 60 + 1 = \boxed{961}.$$

Then, we let $a = 300$ and $b = -1$ to compute 299^2:

$$\begin{aligned} 299^2 &= (300 - 1)^2 \\ &= 300^2 + 2(300)(-1) + (-1)^2 \\ &= 90000 - 600 + 1 \\ &= \boxed{89401}. \end{aligned}$$

3. We can square a number ending in 5 using the following steps:
   1. Erase the 5 at the end of the number to obtain a number k.
   2. Multiply k by $k + 1$ and write the digits 25 at the end of the resulting product.

   For example, we find that $65^2 = 4225$ by evaluating $6 \cdot 7 = 42$, and then placing 25 on the end of this product.

   a. Use this trick to compute 45^2, 95^2, and 115^2.

Solution: We have $4 \cdot 5 = 20$, so we tack 25 on the end of 20 to get $45^2 = \boxed{2025}$. We have $9 \cdot 10 = 90$, so we tack 25 on the end of 90 to find $95^2 = \boxed{9025}$. We have $11 \cdot 12 = 132$, so we tack 25 on the end of 132 to find $115^2 = \boxed{13225}$.

   b. Use the expansion of $(10n + 5)^2$ to explain why this trick works.

Solution: Each number that ends in 5 can be written as $10n + 5$ for some value of n. The number $10n$ results from tacking a 0 onto the end of n, then adding 5 just changes the 0 to a 5. Therefore, when we erase the 5 from the end of $10n + 5$, we are basically reversing these steps to get n back. So, the number that results when we erase the 5 from the end of $10n + 5$ is just n.

When we square $10n + 5$, we have

$$\begin{aligned}(10n+5)^2 &= (10n)^2 + 2(10n)(5) + 5^2 \\ &= 100n^2 + 100n + 25 \\ &= 100(n^2+n) + 25.\end{aligned}$$

The product $100(n^2 + n)$ will end in 00, so when we add on 25, we'll get a number that ends in 25. So, we see where we get the "write the digits 25 at the end of the product" from. But what about that $100(n^2 + n)$? We can factor $n^2 + n$ as $n(n + 1)$, so we have

$$(10n+5)^2 = 100(n^2+n) + 25 = 100n(n+1) + 25.$$

Aha! Now, we see where our steps come from. When we erase the 5 from the end of $10n + 5$, we get the number n. Our rules say to multiply this n by $n + 1$, which gives us $n(n + 1)$. When we tack 25 onto the end of this, we are first multiplying $n(n + 1)$ by 100 (in order to move this number two digits to the left, to make room for the 25), then, we are adding 25. So, tacking 25 onto the end of $n(n + 1)$ gives us $100n(n + 1) + 25$. This number looks familiar! We can reverse our algebra steps above to see that

$$\begin{aligned}100n(n+1) + 25 &= 100n^2 + 100n + 25 \\ &= (10n)^2 + 2(10n)(5) + 5^2 \\ &= (10n+5)^2.\end{aligned}$$

4. Compute $199919981997^2 - 2 \cdot 199919981994^2 + 199919981991^2$

Solution: Instead of squaring those nasty numbers, we let $x = 199919981991$. Then, we can use algebra:

$$\begin{aligned}&199919981997^2 - 2 \cdot 199919981994^2 + 199919981991^2 \\ &= (x+6)^2 - 2(x+3)^2 + x^2 \\ &= x^2 + 12x + 36 - 2(x^2 + 6x + 9) + x^2 \\ &= x^2 + 12x + 36 - 2x^2 - 12x - 18 + x^2 \\ &= \boxed{18}.\end{aligned}$$

That's a lot easier than squaring 12-digit numbers!

5. Factor each of the following expressions:
   a. $t^2 - 49$

Solution: This expression is the difference of two perfect squares, so we can use our difference of squares factorization, $x^2 - y^2 = (x - y)(x + y)$.

This gives

$$\begin{aligned}t^2 - 49 &= t^2 - 7^2 \\ &= \boxed{(t-7)(t+7)}.\end{aligned}$$

   b. $36 - 9x^2$

Solution: Factoring the expression as a difference of squares gives

$$\begin{aligned} 36 - 9x^2 &= 6^2 - (3x)^2 \\ &= (6-3x)(6+3x). \end{aligned}$$

We can then factor a 3 out of each of $6-3x$ and $6+3x$ to give

$$3 \cdot (2-x) \cdot 3 \cdot (2+x) = \boxed{9(2-x)(2+x)}.$$

We could also have factored out a 9 at the beginning:

$$36 - 9x^2 = 9(4-x^2) = 9(2-x)(2+x).$$

c. $121a^2b^4 - c^2$

Solution: Factoring the expression as a difference of squares gives

$$121a^2b^4 - c^2 = (11ab^2)^2 - c^2 = \boxed{(11ab^2 - c)(11ab^2 + c)}.$$

d. $800x^4 - 72x^2y^2$

Solution: At first, it doesn't look like we have a difference of squares factorization because neither term looks like a perfect square. However, we can factor $8x^2$ out of both terms, and then we see the difference of squares:

$$\begin{aligned} &800x^4 - 72x^2y^2 \\ &= 8x^2(100x^2 - 9y^2) \\ &= 8x^2[(10x)^2 - (3y)^2] \\ &= \boxed{8x^2(10x-3y)(10x+3y)}. \end{aligned}$$

6. Given that $55555^2 = 3086358025$, calculate 55556^2 without a calculator.

Solution: In the text, we learned that the difference between $(n+1)^2$ and n^2 is $n + (n+1)$. Therefore, to find 55556^2 given 55555^2, we simply add 55555 and 55556 to 55555^2:

$$\begin{aligned} &55556^2 \\ &= 55555^2 + 55555 + 55556 \\ &= 3086358025 + 111111 \\ &= \boxed{3086469136}. \end{aligned}$$

7. Answer questions below without using a calculator:
   a. Are 3^8 and 2^6 perfect squares?

Solution: Because $(3^4)^2 = 3^8$ and $(2^3)^2 = 2^6$, both 3^8 and 2^6 are perfect squares.

   b. Find the prime factorization of $3^8 - 2^6$. Challenge: See if you can do it in your head!

Solution: Because 3^8 and 2^6 are perfect squares, we can use the difference of squares factorization to find

$$\begin{aligned} 3^8 - 2^6 &= (3^4)^2 - (2^3)^2 \\ &= (3^4 - 2^3)(3^4 + 2^3) \\ &= (81 - 8)(81 + 8) \\ &= \boxed{(73)(89)}. \end{aligned}$$

8. Factor $\omega^4 - 16$ as much as possible.

Solution: Since w^4 and 16 are both perfect squares, we can use our difference of squares factorization:

$$w^4 - 16 = (w^2)^2 - 4^2 = (w^2 - 4)(w^2 + 4).$$

We're not finished! The expression $w^2 - 4$ is also a difference of squares, which we can factor as $w^2 - 4 = (w - 2)(w + 2)$. So, we have

$$w^4 - 16 = (w^2 - 4)(w^2 + 4) = \boxed{(w - 2)(w + 2)(w^2 + 4)}.$$

9. Find all pairs of positive integers m and n such that m^2 is 105 greater than n^2.

$$m^2 = n^2 + 105.$$

It's not at all clear what to do with this equation, but if we subtract n^2 from both sides, we have $m^2 - n^2 = 105$. We like this because we can factor the left side:

$$(m - n)(m + n) = 105.$$

Because m and n are integers, so are $m - n$ and $m + n$. Therefore, both $m - n$ and $m + n$ are factors of 105. We can use the different possible factorizations of 105 to give us the different possible values of $m - n$ and $m + n$:

$$(m - n)(m + n) = 1 \cdot 105 = 3 \cdot 35 = 5 \cdot 21 = 7 \cdot 15.$$

Because m and n are positive, we know that $m - n$ is smaller than $m + n$, so we only have these four cases to consider:

$$\begin{array}{llll} m - n = 1 & m - n = 3 & m - n = 5 & m - n = 7 \\ m + n = 105 & m + n = 35 & m + n = 21 & m + n = 15 \end{array}$$

Each of these systems of equations gives us a solution (m, n). Adding the equations in the first case gives us $2m = 106$, so $m = 53$. Substitution then gives $n = 52$. Similarly, we can work through each of the other three cases to find the four solutions $(m, n) = (53, 52)$; $(19, 16)$; $(13, 8)$; $(11, 4)$. □